

Salifort Motors

Employee Retention Project

ISSUE / PROBLEM

Salifort Motors seeks to improve employee retention and answer the following question:

What's likely to make the employee leave the company?

RESPONSE

We analyzed company data and ran statistical tests that confirmed overwork drives turnover.

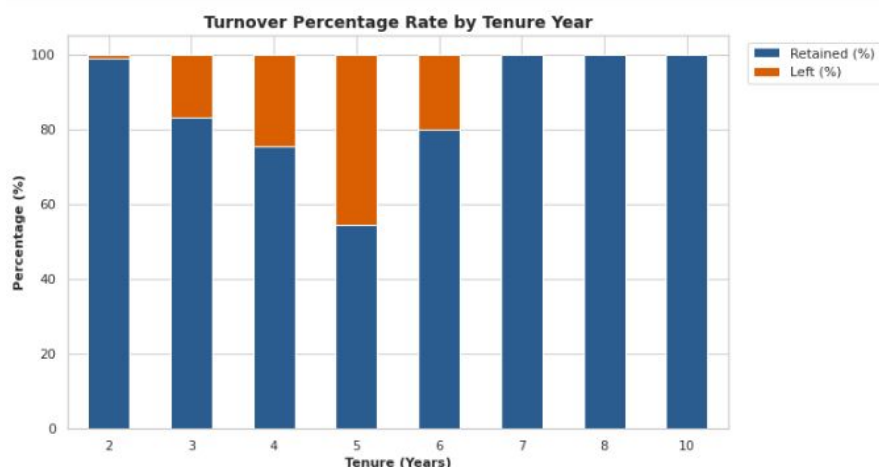
We built machine learning models to predict which employees are most likely to leave.

The Random Forest model worked best, accurately identifying high-risk employees so HR can help them early.

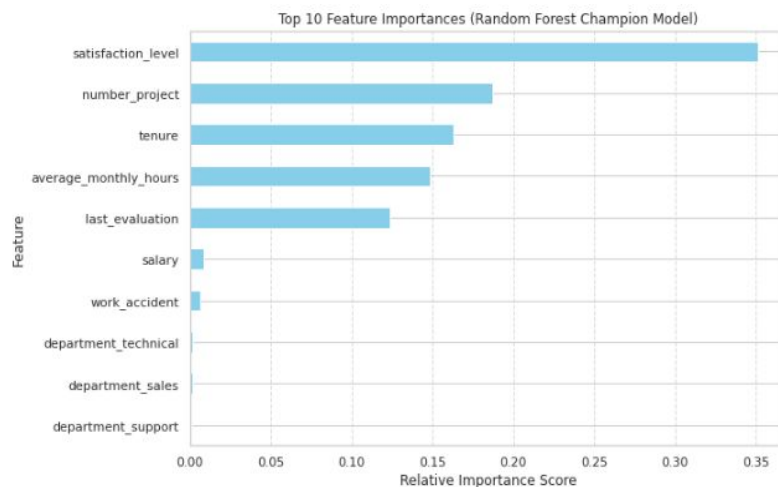
IMPACT

Managers can now spot flight risks before employees hand in their resignation.

Fixing workload issues will save the company money on hiring and training new staff.



Barplot above shows the most relevant variables: `'last_evaluation'`, `'number_project'`, `'tenure'` and `'overworked'`.



In the random forest model above, `'last_evaluation'`, `'number_project'`, `'tenure'`, `'average_monthly_hours'`, `'last_evaluation'`, `'salary'`, and `'work_accident'` have the highest importance. These variables are most helpful in predicting the outcome variable, `'left'`.

INSIGHTS/NEXT STEPS

- **Cap monthly hours:** Limit working hours to 200 hours per month and assign 3 to 5 projects per person.
- **Target 3-to-5 year staff:** Offer promotions and pay raises to employees reaching their 3rd year to stop the turnover peak.
- **Retrain without leakage data:** Test the model without `satisfaction_level` and `last_evaluation` to make sure it predicts turnover well in real-world deployment.
- **Keep HR actions supportive:** Use predictions only to help overworked staff, never to penalize or target them.